Solutions to Tutorial 4 problems

(1)

Here $f = y'^2 + yy' + y^2$

The integral $\int_0^P f \, dx$ is stationary if

f satisfies the so-called Euler-Lagrange equation:

$$\frac{\partial f}{\partial y} = \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) \quad ; \quad y' = \frac{dy}{dx}$$

$$\Rightarrow \quad y' + 2y = 2y'' + y'$$

$$\Rightarrow \quad y'' = y$$

The general solution of this equation is:

$$y(x) = A \sinh x + B \cosh x$$

Given: $y(0) = 0, \quad y(1) = 1$

So, we see that

$$B = 0, \qquad A = \frac{1}{\sinh(1)}$$

The required solution is:

$$y(x) = \frac{\sinh(x)}{\sinh(1)} \qquad //$$

(2)

(a) $$f = \frac{\dot{x}^2}{4t}$$

$$\frac{\partial f}{\partial x} = 0, \qquad \frac{\partial f}{\partial \dot{x}} = \frac{\dot{x}}{2t}$$

Euler - Lagrange equation takes the form

$$\frac{d}{dt}\left(\frac{\partial f}{\partial \dot{x}}\right) = 0$$

$$\Rightarrow \frac{d}{dt}\left(\frac{\dot{x}}{2t}\right) = 0$$

$\Rightarrow \dfrac{\dot{x}}{2t} = c$, a constant of integration

$\Rightarrow x = ct^2 + d$, d is a constant of integration

$$\left.\begin{array}{l} x(1) = 5 \Rightarrow 5 = c + d \\ x(2) = 11 \Rightarrow 11 = 4c + d \end{array}\right\} \Rightarrow \begin{array}{l} c = 2 \\ d = 3 \end{array}$$

Hence: $$x = 2t^2 + 3$$ //

(b) Here, $$f = 2x \sin t - \dot{x}^2$$

$$\frac{\partial f}{\partial x} = 2 \sin t \; ; \quad \frac{\partial f}{\partial \dot{x}} = -2\dot{x}$$

The EL eq^n is

$$\frac{d}{dt}\left(\frac{\partial f}{\partial \dot{x}}\right) - \frac{\partial f}{\partial x} = 0$$

$$\Rightarrow \quad \frac{d}{dt}\left(-2\dot{x}\right) - 2\sin t = 0$$

$$\Rightarrow \quad \ddot{x} = -\sin t$$

The general solution of this equation is

$$x = \sin t + \alpha t + \beta$$

where α and β are integration constants.

$$x(0) = 0 \Rightarrow \beta = 0$$

$$x(\pi) = 0 \Rightarrow \alpha = 0$$

So we have:

$$x = \sin t$$

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(3)

$$f = f(y, y', x) \quad ; \quad \text{here } y' = \frac{dy}{dx}$$

Now,

$$\frac{df}{dx} = \frac{\partial f}{\partial y}\frac{dy}{dx} + \frac{\partial f}{\partial y'}\frac{dy'}{dx} + \frac{\partial f}{\partial x}$$

$$= y'\frac{\partial f}{\partial y} + y''\frac{\partial f}{\partial y'} + \frac{\partial f}{\partial x} \quad \rightarrow (1)$$

Again,

$$\frac{d}{dx}\left(y'\frac{\partial f}{\partial y'}\right) = y''\frac{\partial f}{\partial y'} + y'\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) \quad \rightarrow (2)$$

From Eq. (1) let us substitute for $y''\frac{\partial f}{\partial y'}$ to obtain:

$$\frac{d}{dx}\left(y'\frac{\partial f}{\partial y'}\right) = \frac{df}{dx} - \frac{\partial f}{\partial x} - y'\frac{\partial f}{\partial y} + y'\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right)$$

$$= \frac{df}{dx} - \frac{\partial f}{\partial x} + y'\left(\frac{d}{dx}\frac{\partial f}{\partial y'} - \frac{\partial f}{\partial y}\right) \quad \rightarrow (3)$$

Using Euler-Lagrange eq^n,

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0 \quad \text{in } (3)$$

we obtain:

$$\frac{d}{dx}\left(y' \frac{\partial f}{\partial y'}\right) = \frac{df}{dx} - \frac{\partial f}{\partial x}$$

$$\Rightarrow \boxed{\frac{\partial f}{\partial x} - \frac{d}{dx}\left(f - y'\frac{\partial f}{\partial y'}\right) = 0}$$

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(4) In plane polar coordinates, the length element ds is given by

$$ds^2 = dr^2 + r^2 d\theta^2$$

$$\Rightarrow ds = \left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]^{1/2} d\theta$$

The Fermat time functional

$$T[r] = \int n\left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]^{1/2} d\theta$$

is stationary, where $n = n(r)$ is the refractive index.

Recall, Fermat principle requires, time

$$T = \int_P^Q \frac{ds}{c/n}$$

$$= \frac{1}{c}\int n\, ds,$$

to be minimum

Here :

We have, $f = r\left(r^2 + \dot{r}^2\right)^{1/2}$

with no explicit θ dependence.

So, from the second form of E-L equation (refer to eq^n (1) on the Tutorial 4 problem-set) :

$$\dot{r}\frac{\partial f}{\partial \dot{r}} - f = \text{constant} = c_1 \text{, say}$$

$$\Rightarrow \frac{r\dot{r}^2}{\left(r^2+\dot{r}^2\right)^{1/2}} - r\left(r^2+\dot{r}^2\right)^{1/2} = c_1$$

$$\Rightarrow \frac{r\dot{r}^2 - rr^2 - r\dot{r}^2}{\left(r^2+\dot{r}^2\right)^{1/2}} = c_1$$

$$\Rightarrow \boxed{\frac{rr^2}{\left(r^2+\dot{r}^2\right)^{1/2}} = -c_1 = b \text{, a constant}} \rightarrow (A)$$

$$\Rightarrow r^2r^4 = b^2\left(r^2+\dot{r}^2\right)$$

$$\Rightarrow b^2\dot{r}^2 = r^2r^4 - b^2r^2$$

$$\Rightarrow \dot{r}^2 = r^2\left(\frac{r^2r^2}{b^2} - 1\right)$$

Second part of the problem:

Write: $\dot{r} = r \tan\psi$

$$\tan\psi = \frac{1}{r}\frac{dr}{d\theta}$$

On replacing $\dot{r}$ in eqn (A) by

$r \tan\psi$:

$$\frac{nr^2}{\left(r^2 + r^2\tan^2\psi\right)^{1/2}} = c$$

$$\Rightarrow \quad \frac{nr}{\sqrt{1+\tan^2\psi}} = c$$

$$\Rightarrow \quad \boxed{nr \cos\psi = \text{constant}}$$

(5) $$L = \dot{q}^2 - 4q^2$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0$$

$$\Rightarrow \quad \ddot{q} + 4q = 0$$

This is the classical SHM equation with $\omega = 2$

Clearly $q = \sin 2t$ is a possible motion of the system. //

The action S for the time interval $[0, \tau]$ is

$$S[q] = \int_0^{\tau} \left(\dot{q}^2 - 4q^2 \right) dt$$

By the principle of least action the motion $q_s = \sin 2t$ makes S stationary. Let's prove it from the first principle.

Consider the function

$$q = q_s + \delta q \quad , \text{ where } \delta q \text{ is}$$

a small variation such that $\delta q(0) = 0 = \delta q(\tau)$.

Then,

$$S[q_s + \delta q] = \int_0^{\tau} \left[\left(\dot{q}_s + \delta \dot{q} \right)^2 - 4 \left(q_s + \delta q \right)^2 \right] dt$$

$$= \int_0^{\tau} \left[\left(2\cos 2t + \delta \dot{q} \right)^2 - 4 \left(\sin 2t + \delta q \right)^2 \right] dt$$

$$= \int_0^\tau \Big[4\cos 4t + 4\,\delta\dot{q}\cos 2t - 8\,\delta q \sin 2t + \delta\dot{q}^2 - 4\,\delta q^2 \Big]$$

$$= \sin 4\tau + 4\,\delta q \cos 2t \Big|_0^\tau + \int_0^\tau \left[\delta\dot{q}^2 - 4\,\delta q^2 \right] dt$$

$$= \sin 4\tau + 0 + \int_0^\tau \left(\delta\dot{q}^2 - 4\,\delta q^2 \right) dt$$

Taking $\delta q \equiv 0$

$$S[q_s] = \sin 4\tau$$

Hence

$$S[q_s + \delta q] = S[q_s] + \int_0^\tau \left(\delta\dot{q}^2 - 4\,\delta q^2 \right) dt$$

$$= S[q_s] + \underbrace{\mathcal{O}\left(\|\delta q\|^2 \right)}_{\text{of order } \|\delta q\|^2}$$

Thus, in accordance with the principle of least action, the motion $q_s = \sin 2t$ makes the action functional stationary.